

UNIFIED MENTOR

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Batch	25Mar thru 25Jun 2026
Type of Intern	Data Analytics Internship – Healthcare Analytics
Project Title	1. System Capacity & Care Load Analytics for Unaccompanied Children US – Human Health Services
Github Repo Link	https://github.com/M339KUMAR/DataAnalytics1/
Project Pdf Link	
Video Link	https://github.com/M339KUMAR/DataAnalytics1/blob/main/P1%20-%20Healthcare%20Analytics%2020260607_142520.mp4
Streamlit Project Link	https://dataanalytics1-umdaushhsproject1.streamlit.app/

Abstract :-

The Unaccompanied Alien Children (UAC) Program is a federally mandated initiative under which children apprehended by US Customs & Border Protection CBP are transferred to Dept of HHS, Health & Human Services for medical screening, quarantine, Sheltering & eventual placement with vetted sponsors. The Program functions as a dynamic care pipeline involving Intake, Transfer, and Medical Psychological & Welfare Support with Discharge reunification with sponsors.

Problem Statement:-

Although Daily Operational data is collected HHS currently lacks a centralised analytical framework to continuously assess total care system load, balance between inflow & outflow, capacity Stress & relief periods sustainability of Care Delivery overtime without structured analytics, decision making becomes proactive increasing the risk of overcrowding, delayed care & strain on healthcare infrastructure.

Dataset Description:-

The dataset has 6 Variables & 720 Rows/Records,

- 1 Date
- 2 Children apprehended and placed in CBP custody*
- 3 Children in CBP custody
- 4 Children transferred out of CBP custody
- 5 Children in HHS Care
- 6 Children discharged from HHS Care

Objectives:-

- Quantify daily & cumulative care load across CBP & HHS
- Identify periods of capacity strain & relief
- Analyse balance between intake, transfers & discharges
- Support healthcare staffing & shelter planning
- Improve situational awareness of policy makers
- Design a system for data driven Humanitarian response

Methodology:

1. Data Ingestion & Structuring,
2. Derived Healthcare Metrics,
3. data Quality & Validation,
4. Pressure & Stress Identification

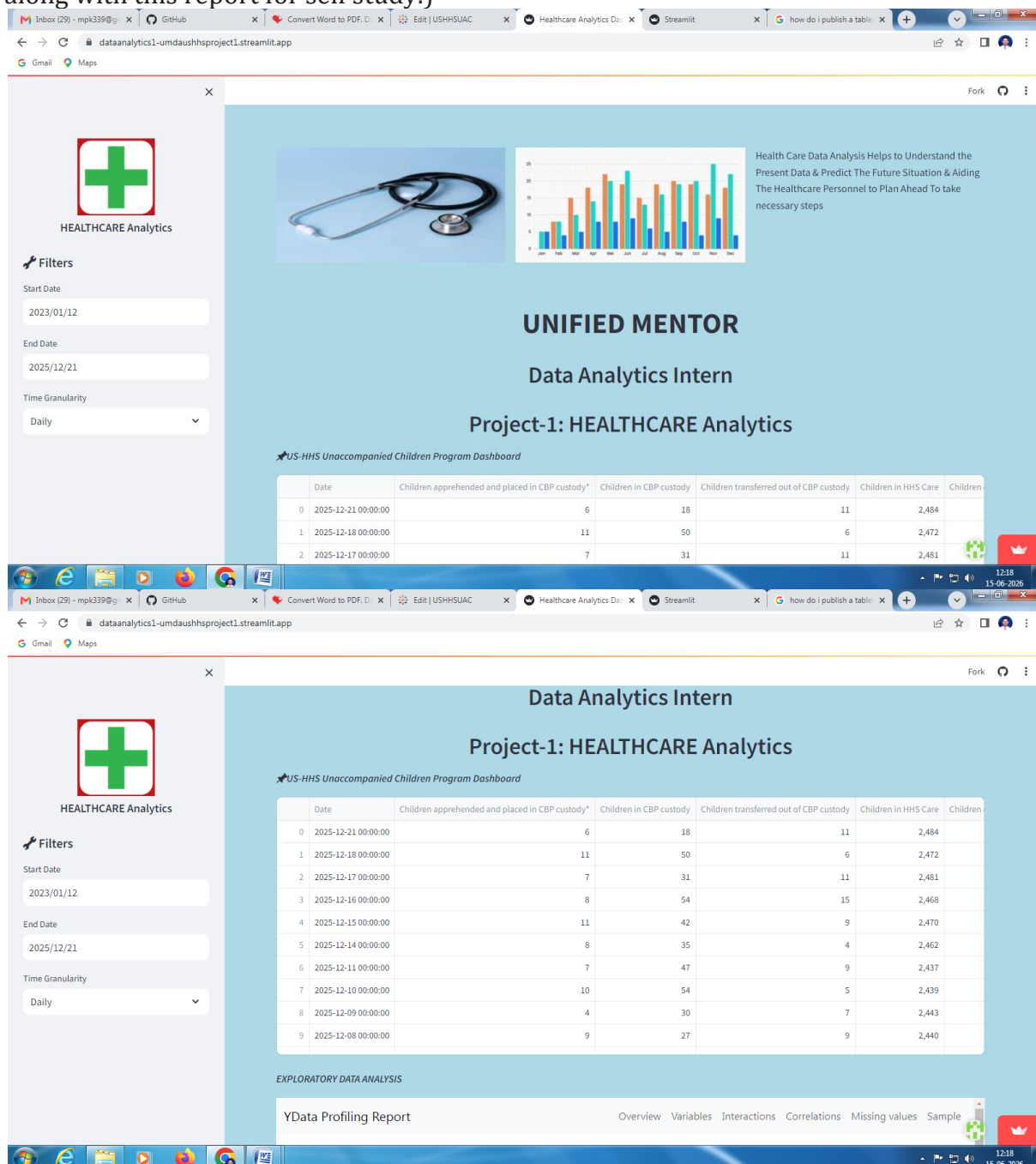
System Design:

The Existing system in US HHS Dept the UAC Program needs improvement to predict the Total Care Load of the System in advance to take the necessary steps like Staffing to accomodate the increase or decrease in the care load, This project aims to build a Dashboard to Predict the Total Care load using a Machine Learning Time Series Forecasting Technique(SARIMA), & also perform Exploratory Data Analysis(EDA) on the data to understand the variables, The user inputs the data, and predict the Total system Care load capacity, The Key Performance Indicators Metrics, are also Displayed for helping HHS personnel to understand the unforeseen Demand.

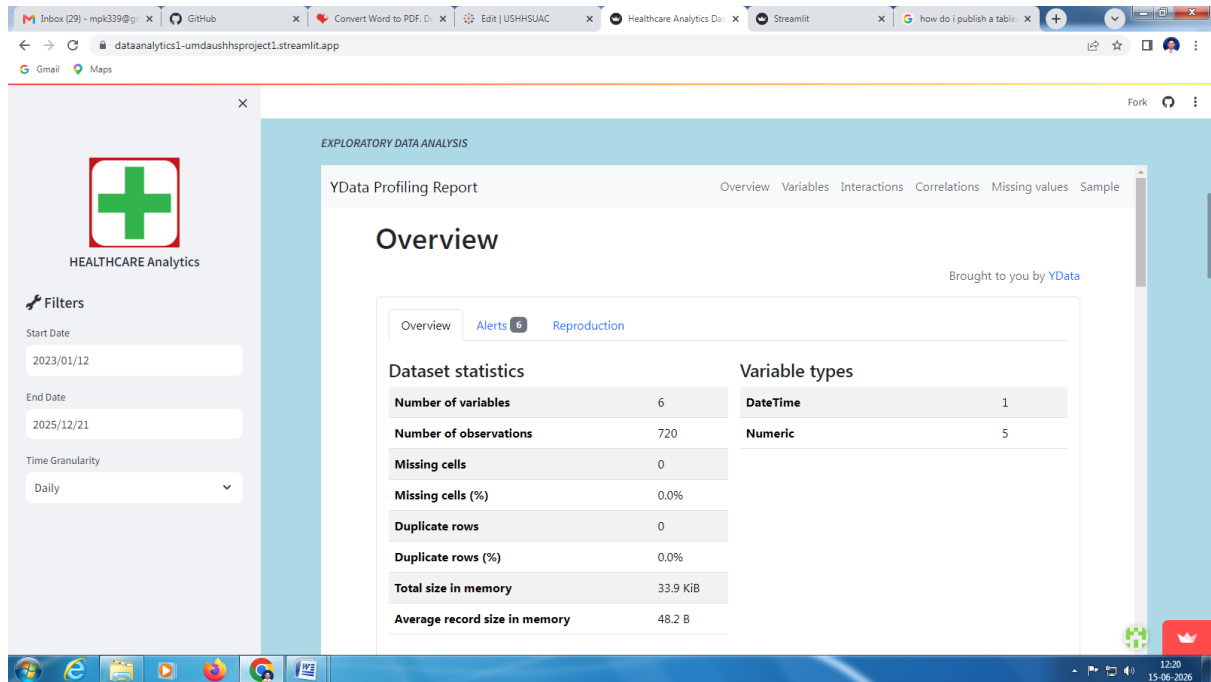
Conclusion & Recommendation:

The Present Report tries to make an attempt to aid the HHS Dept personnel, but do not replace the current system, it complements the existing rather than completely replacing it, However the new system helps to identify the key performance metrics and so on .

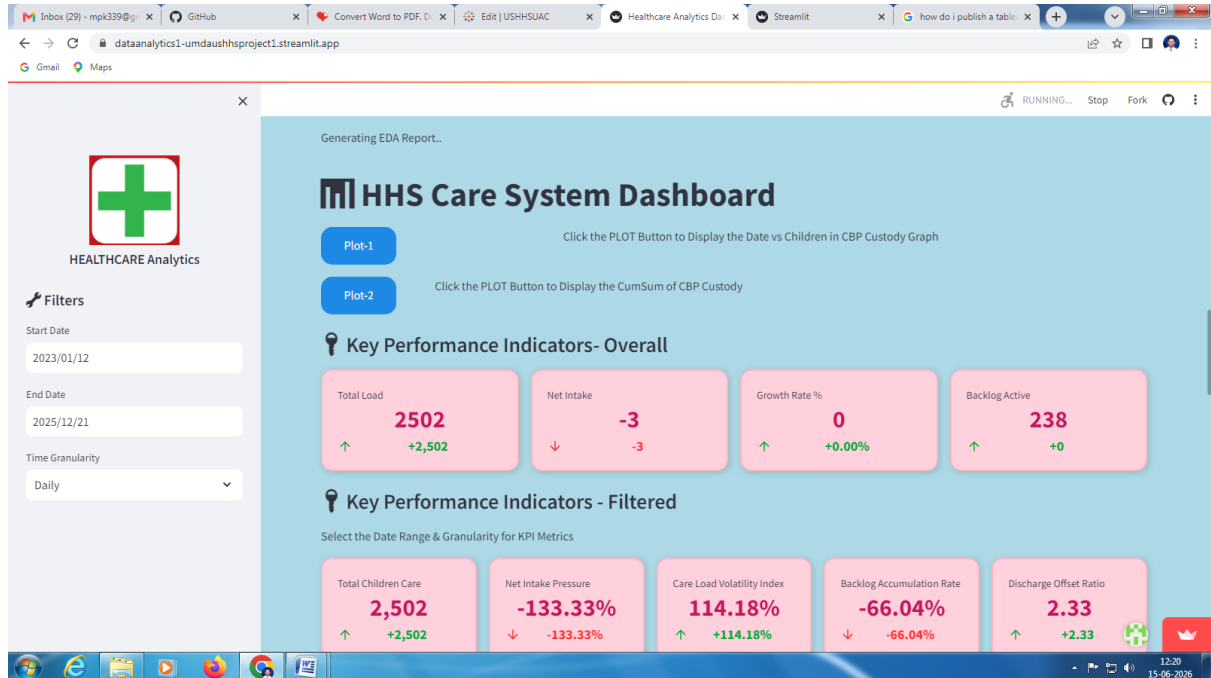
(In order to make more visually appealing, screenshots of the system are also presented along with this report for self study.)



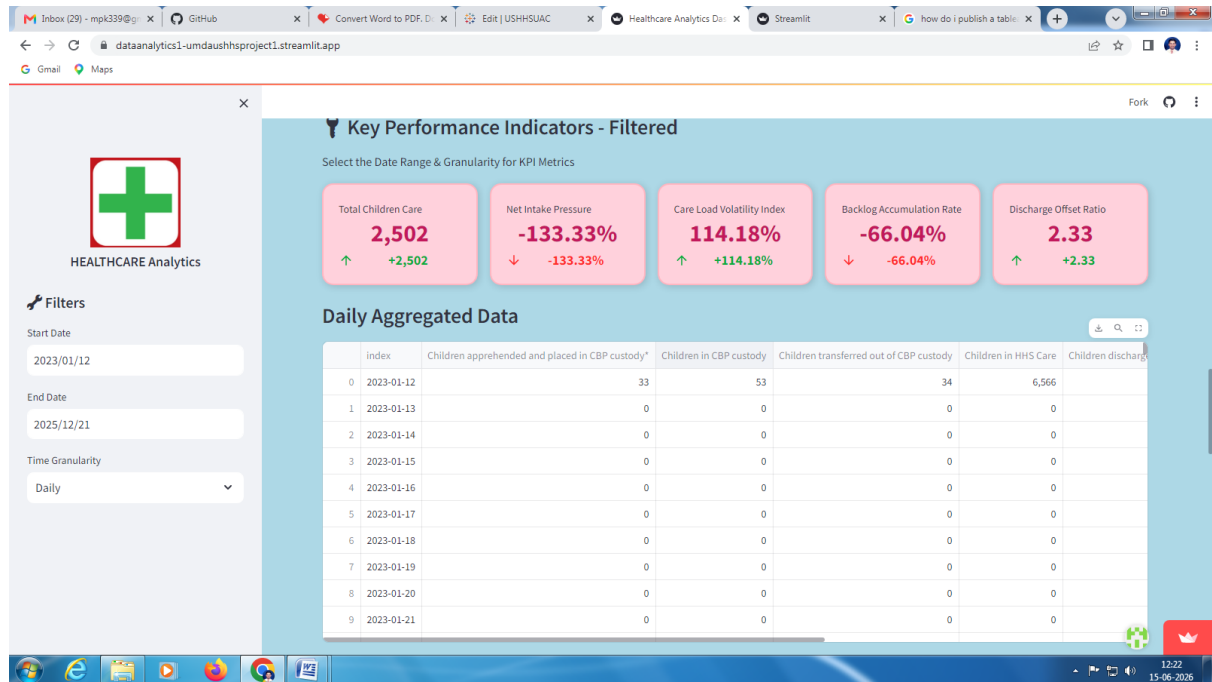
Exploratory Data Analysis



HHS Care System Dashboard



Key Performance Indicators - Overall



Machine Learning Model

Machine Learning Model Predictions

★Note: All fields are Mandatory, pls enter your values

To Forecast Total Load for Next Day

1.Children apprehended and placed in CBP custody (range:1-333)

0

2.Children in CBP custody (range:7-531)

0

3.Children transferred out of CBP custody (range:0-440)

0

4.Children in HHS Care (range:1972-11516)

0

5.Children discharged from HHS Care (range:0-505)

0

After Entering Values - Use The Train Model Button First

Train Model

Click The Predict Button to forecast for next day

Contd...

The screenshot shows the initial interface of the 'HEALTHCARE Analytics' Streamlit application. On the left, there is a sidebar with a green cross logo and the text 'HEALTHCARE Analytics'. Below the logo, there is a 'Filters' section with input fields for 'Start Date' (2023/01/12), 'End Date' (2025/12/21), and a 'Time Granularity' dropdown menu set to 'Daily'. The main area of the app has a title '5 Children discharged from HHS Care (range:0-505)' and a slider set to 0. Below this, there are instructions: 'After Entering Values - Use The Train Model Button First', a 'Train Model' button, and 'Click The Predict Button to forecast for next day', followed by a 'Predict1' button. A section titled 'To Forecast Total Load within next 7 days(Week)' contains the instruction 'Use the slider to set num. of days to forecast & Enter The Values'. Below this is a slider for 'Select future prediction steps' set to 3. At the bottom, there is a table with 6 columns: 'Children apprehended and placed in CBP custody*', 'Children in CBP custody', 'Children transferred out of CBP custody', 'Children in HHS Care', and 'Children'. The table has 3 rows of data, all with values of 0.

	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0

This screenshot shows the 'HEALTHCARE Analytics' app after clicking the 'Predict1' button. The 'Train Model' button is now disabled. The 'Predict1' button has been replaced by a 'Predict2' button. Below the 'Predict2' button, there is a section titled '*Tableau Dashboard' which is currently empty. The rest of the interface, including the filters and the data table, remains the same as in the previous screenshot.

This screenshot shows the 'HEALTHCARE Analytics' app after clicking the 'Predict2' button. The 'Predict2' button is now disabled. Below it, the '*Tableau Dashboard' section is populated with the following text: 'PRAVEENKUMAR MOPURU', 'UMID20032685175', 'Data Analyst Intern', and '25th March 2026 Batch'. The rest of the interface, including the filters and the data table, remains the same as in the previous screenshots.

THANK YOU